Centralized, Decentralized, and Hybrid Connectivity Control for Multi-Robot Systems using Predicted Control Barrier Functions

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Abstract

This report presents centralized, decentralized, and hybrid global–local control strategies for multi-robot systems with double-integrator dynamics based on predicted Control Barrier Functions (CBFs).

The centralized approach computes joint accelerations for all agents using a single quadratic program (QP), enforcing collision avoidance, obstacle avoidance, and connectivity at the network level.

The decentralized approach distributes computation across agents, each of which solves a local QP using only neighbor information. This greatly improves scalability while retaining safety and practical connectivity.

Finally, a hybrid global–local design augments the decentralized controller with a connectivity feedback term driven by the algebraic connectivity λ_2 (or its distributed estimate). The global signal modulates local connectivity CBF gains, enabling the swarm to react collectively when global connectivity becomes weak.

Contents

Chapter 1

Introduction

Multi-robot systems require coordination strategies that guarantee safety, connectivity, and convergence towards shared goals. Control Barrier Functions (CBFs) provide a powerful framework for enforcing safety-type constraints while allowing flexible motion control.

In this report, we consider a team of N robots in the plane. Each robot is modeled as a double integrator, and its acceleration input is chosen by solving a QP that trades off goal tracking against safety and connectivity constraints.

We first describe a centralized CBF-based controller, then a decentralized version, and finally a hybrid global–local connectivity scheme that introduces a λ_2-based global feedback into the decentralized CBF design.

Chapter 2

System Dynamics and Preliminaries

2.1 Double-integrator model

Each robot $i \in \{1, \ldots, N\}$ is modeled as:

$$\dot{x}_i = v_i, \qquad \dot{v}_i = u_i, \tag{2.1}$$

where $x_i, v_i, u_i \in \mathbb{R}^2$ denote the position, velocity, and acceleration of agent i respectively. We denote by $x = [x_1^\top, \ldots, x_N^\top]^\top$ and $v = [v_1^\top, \ldots, v_N^\top]^\top$ the stacked state vectors.

2.2 Nominal goal-seeking controller

A simple nominal controller for each agent is a PD law driving the robot toward a goal position x_{goal}:

$$u_{\text{nom},i} = -k_p(x_i - x_{\text{goal}}) - k_d v_i, \tag{2.2}$$

with $k_p > 0$ and $k_d > 0$. The QP-based controller will then minimally modify $u_{\text{nom},i}$ to satisfy CBF constraints.

2.3 Predicted-distance Control Barrier Functions

We use predicted-distance CBFs that depend on both position and velocity through a finite prediction horizon $T_{\text{pred}} > 0$.

For a pair of agents i and j, define the relative position and velocity:

$$x_{ij} = x_i - x_j, \qquad v_{ij} = v_i - v_j. \tag{2.3}$$

We define a predicted relative position:

$$x_{ij}^T = x_{ij} + T_{\text{pred}} v_{ij}. \tag{2.4}$$

A generic barrier function $h(x, v)$ is associated to a safety condition ($h(x, v) \geq 0$). The CBF condition has the form:

$$\dot{h}(x, v, u) + \alpha h(x, v) \geq 0, \tag{2.5}$$

where $\alpha > 0$ is typically a scalar gain. When $\dot{h}$ is affine in the control input u, this yields linear inequalities in the QP.

2.3.1 Collision avoidance CBF

To enforce a minimum distance $d_{\min}$ between agents i and j, we define:

$$h_{\text{col}}^{ij}(x, v) = \left\| x_{ij} + T_{\text{pred}} v_{ij} \right\|^2 - d_{\min}^2. \tag{2.6}$$

The associated CBF constraint is:

$$\dot{h}_{\text{col}}^{ij}(x, v, u) + \alpha_{\text{col}} h_{\text{col}}^{ij}(x, v) \geq 0. \tag{2.7}$$

2.3.2 Connectivity CBF

We also want to avoid losing communication links. Let $R_{\max}$ be the maximum communication range. A predicted connectivity CBF is:

$$h_{\text{conn}}^{ij}(x, v) = R_{\max}^2 - \left\| x_{ij} + T_{\text{pred}} v_{ij} \right\|^2. \tag{2.8}$$

The CBF inequality becomes:

$$\dot{h}_{\text{conn}}^{ij}(x, v, u) + \alpha_{\text{conn}} h_{\text{conn}}^{ij}(x, v) \geq 0. \tag{2.9}$$

This keeps neighboring agents from drifting outside communication range.

2.3.3 Obstacle avoidance

For a static obstacle centered at x_{obs} with radius r_{obs} and safety margin r_{safe}, we define:

$$h_{\text{obs}}^{i}(x, v) = \left\| x_i + T_{\text{pred}} v_i - x_{\text{obs}} \right\|^2 - (r_{\text{obs}} + r_{\text{safe}})^2. \tag{2.10}$$

The corresponding CBF constraint is:

$$\dot{h}^i_{\mathrm{obs}}(x, v, u_i) + \alpha_{\mathrm{obs}} h^i_{\mathrm{obs}}(x, v) \geq 0. \tag{2.11}$$

Chapter 3

Centralized Predicted CBF Controller

In the centralized implementation (e.g., script `cbf_centralized.m`), a single QP is solved that includes all agents' accelerations as decision variables.

3.1 Centralized QP formulation

Stack all controls into $u = [u_1^\top, \ldots, u_N^\top]^\top$. The QP is:

$$\begin{aligned} \min_{u} \quad & \|u - u_{\text{nom}}\|^2 \\ \text{s.t.} \quad & \dot{h}_{\text{col}}^{ij}(x, v, u) + \alpha_{\text{col}} h_{\text{col}}^{ij}(x, v) \geq 0, \\ & \dot{h}_{\text{conn}}^{ij}(x, v, u) + \alpha_{\text{conn}} h_{\text{conn}}^{ij}(x, v) \geq 0, \\ & \dot{h}_{\text{obs}}^{i}(x, v, u) + \alpha_{\text{obs}} h_{\text{obs}}^{i}(x, v) \geq 0, \end{aligned} \tag{3.1}$$

for all relevant agent pairs (i, j) and obstacles.

This global QP enforces all constraints simultaneously with consistent coupling among all robots.

3.2 Architecture illustration

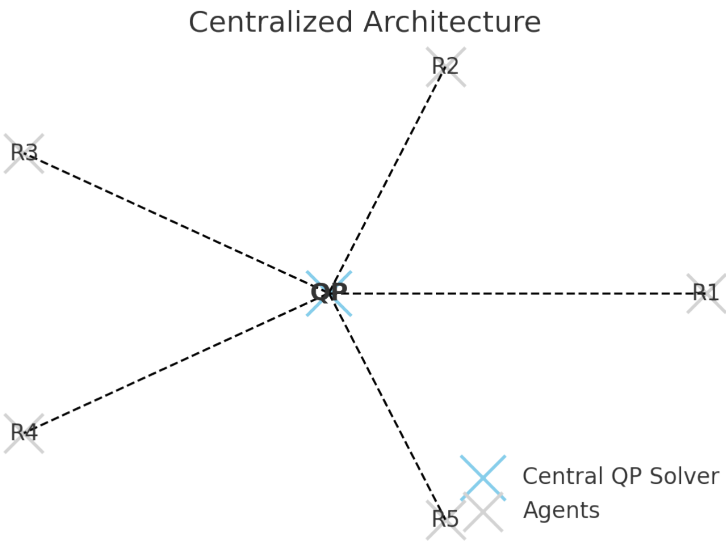

Figure 3.1: Centralized architecture: a single QP solver computes the joint accelerations for all agents.

3.3 Discussion

The centralized approach yields globally consistent safety and connectivity, and is conceptually straightforward. However, the size of the QP grows with N, both in the number of decision variables ($2N$ in 2D) and in the number of pairwise constraints (on the order of N^2). This limits scalability.

Moreover, using λ_2 directly in a centralized CBF would require computing or approximating $\partial\lambda_2/\partial x_i$, which is computationally expensive and less amenable to real-time implementation.

Chapter 4

Decentralized Predicted CBF Controller

In the decentralized implementation (`cbf_decentralized.m`), each agent solves a small QP using only local information from neighbors and obstacles.

4.1 Local neighborhood and sensing

For each agent i, a neighborhood $\mathcal{N}_i$ is defined as the set of agents within a communication or sensing radius R_{comm}. Each robot has access to x_j, v_j for $j \in \mathcal{N}_i$.

4.2 Local QP for agent i

The local decision variable is $u_i \in \mathbb{R}^2$. The local QP is:

$$\begin{aligned}
\min_{u_i} \quad & \|u_i - u_{\text{nom},i}\|^2 \\
\text{s.t.} \quad & \dot{h}_{\text{col}}^{ij}(x, v, u_i) + \alpha_{\text{col}} h_{\text{col}}^{ij}(x, v) \geq 0, \quad \forall j \in \mathcal{N}_i, \\
& \dot{h}_{\text{conn}}^{ij}(x, v, u_i) + \alpha_{\text{conn}} h_{\text{conn}}^{ij}(x, v) \geq 0, \\
& \dot{h}_{\text{obs}}^{i}(x, v, u_i) + \alpha_{\text{obs}} h_{\text{obs}}^{i}(x, v) \geq 0.
\end{aligned} \tag{4.1}$$

Here, the influence of neighbors' inputs u_j is typically approximated via zero-order hold (using their previous values).

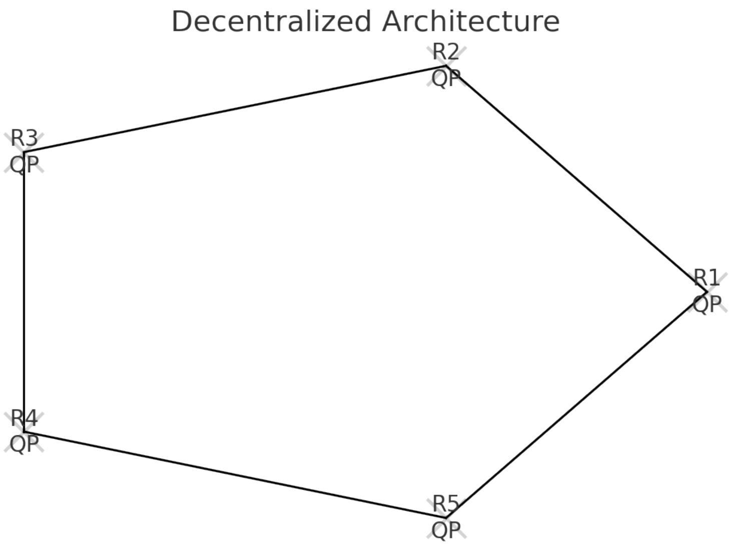

Figure 4.1: Decentralized architecture: each agent solves its own local QP using neighbor and obstacle information.

4.3 Adding global awareness via a distributed λ_2 estimate

A purely local decentralized scheme may (in principle) lose global connectivity even if all local connectivity CBFs are satisfied. To counter this, one can introduce a global connectivity indicator based on the algebraic connectivity λ_2 of the communication graph Laplacian:

$$L = D - A, \qquad \lambda_2(L) \geq 0, \tag{4.2}$$

where A is the adjacency matrix and D is the degree matrix.

In practice, a distributed estimator can provide each agent with an estimate $\hat{\lambda}_2$. In the hybrid design (described in the next chapter), this estimate is used to modulate local connectivity CBF gains.

Chapter 5

Hybrid Global–Local Connectivity Enforcement

This chapter describes the hybrid global–local controller implemented in the script `cbf_hybrid.m`. It combines local predicted CBFs with a global connectivity feedback based on the algebraic connectivity λ_2.

5.1 Motivation

Purely local connectivity CBFs enforce that neighboring robots remain within range, but they do not directly enforce a global property such as $\lambda_2(L) > \varepsilon$. A global connectivity indicator can reveal when the overall communication graph is becoming weak, even if local distances are still acceptable.

The idea is to:

- use a distributed algorithm (or, in simulation, the true Laplacian) to estimate $\hat{\lambda}_2$, and
- modulate the local connectivity CBF gains based on $\hat{\lambda}_2$.

5.2 Global connectivity feedback

Let $\lambda_2(t)$ be the second-smallest eigenvalue of the communication graph Laplacian at time t (the algebraic connectivity). In a real system, each agent would have an estimate $\hat{\lambda}_2(t)$ obtained from a distributed estimator.

We define:

$$h_\lambda = \lambda_2 - \varepsilon_\lambda, \tag{5.1}$$

where $\varepsilon_\lambda > 0$ is a desired lower bound for connectivity.

Instead of directly enforcing a CBF on h_λ, we use h_λ to modulate the local connectivity CBF gains. Define a global gain factor γ_{glob}:

$$\gamma_{\text{glob}}(\hat{\lambda}_2) = \begin{cases} 1 + k_\lambda(\lambda_{\text{warn}} - \hat{\lambda}_2), & \hat{\lambda}_2 < \lambda_{\text{warn}}, \\ 1, & \hat{\lambda}_2 \geq \lambda_{\text{warn}}, \end{cases} \tag{5.2}$$

saturated at a maximum value $\gamma_{\max}$.

The effective connectivity CBF gain becomes:

$$k_{\text{conn,eff}} = \gamma_{\text{glob}}\, k_{\text{conn}}. \tag{5.3}$$

When $\hat{\lambda}_2$ decreases toward ε_λ, the factor γ_{glob} increases, making connectivity constraints more stringent across the entire network.

5.3 Activation logic

To avoid violent initial transients, the global feedback is activated only after the system reaches a sufficiently connected configuration. The typical logic is:

- Initially, $\gamma_{\text{glob}} = 1$ and only local CBFs are active.
- When $\hat{\lambda}_2(t)$ first satisfies $\hat{\lambda}_2(t) \geq \varepsilon_\lambda$, the global feedback is *enabled* permanently.
- For t beyond that point, if $\hat{\lambda}_2(t) < \lambda_{\text{warn}}$, then γ_{glob} increases above 1, reinforcing local connectivity CBFs.

This avoids large impulsive behaviors at startup when the network might be poorly connected. Once connectivity is achieved, the hybrid controller works to keep it from degrading.

5.4 Local QP with modulated connectivity gains

The local QP structure remains the same as in the purely decentralized case, but α_{conn} (or k_{conn}) is replaced by the effective gain $k_{\text{conn,eff}}$:

$$\begin{aligned} \min_{u_i} \quad & \|u_i - u_{\text{nom},i}\|^2 \\ \text{s.t.} \quad & \dot{h}^{ij}_{\text{col}}(x, v, u_i) + \alpha_{\text{col}} h^{ij}_{\text{col}}(x, v) \geq 0, \\ & \dot{h}^{ij}_{\text{conn}}(x, v, u_i) + k_{\text{conn,eff}} h^{ij}_{\text{conn}}(x, v) \geq 0, \\ & \dot{h}^{i}_{\text{obs}}(x, v, u_i) + \alpha_{\text{obs}} h^{i}_{\text{obs}}(x, v) \geq 0. \end{aligned} \tag{5.4}$$

Hence, when global connectivity weakens (small $\hat{\lambda}_2$), all agents tighten their connectivity constraints in a coherent manner, promoting a swarm-level reaction.

5.5 Block diagram illustration

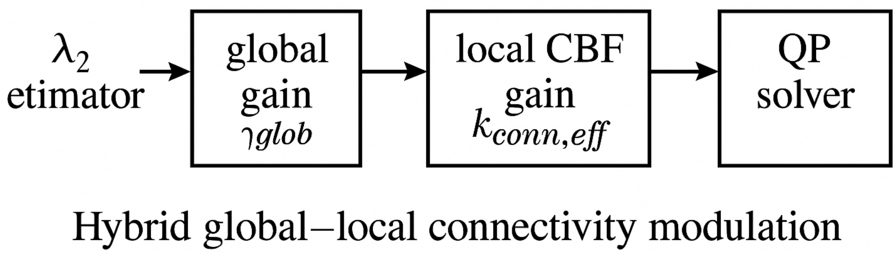

Figure 5.1: Hybrid global–local connectivity modulation. The distributed $\hat{\lambda}_2$ estimate feeds a global gain γ_{glob}, modulating the local connectivity CBF gain before entering each agent's QP.

5.6 Stress-test scenario with a special agent

In the script `cbf_hybrid.m`, a "special" agent is temporarily assigned a different goal position after global connectivity has been established (i.e., once $\hat{\lambda}_2(t) \geq \varepsilon_\lambda$). This creates a scenario where one robot attempts to move away from the formation.

Local collision and connectivity CBFs work to keep the special agent from breaking links, while the global factor γ_{glob} increases when $\hat{\lambda}_2$ drops, reinforcing connectivity constraints for the entire network. This makes the effect of global connectivity feedback clearly visible in simulation.

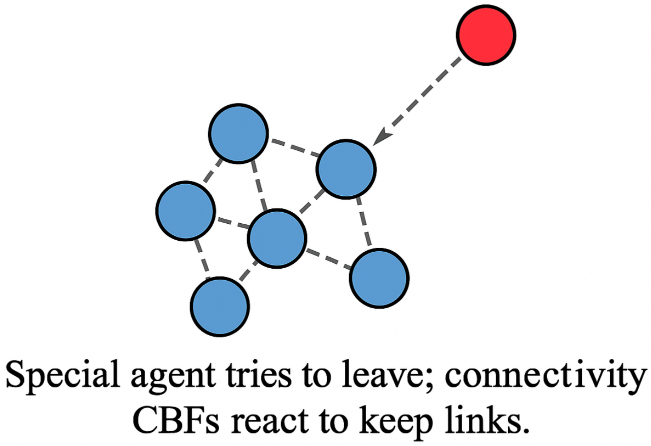

Figure 5.2: Illustrative multi-agent scenario: the special agent tries to leave the group, while hybrid global–local connectivity CBFs act to preserve links.

5.7 Discussion

The hybrid approach preserves the scalability of the decentralized scheme while introducing a principled global connectivity signal. It does not impose a hard CBF directly on λ_2, which would require gradients of eigenvalues with respect to agent positions. Instead, it uses $\hat{\lambda}_2$ to shape the local CBF gains, yielding a smooth and distributed mechanism for enhancing connectivity.

One limitation is that the hybrid scheme does not *guarantee* that λ_2 stays above ε_λ in all cases, especially if task demands (e.g., conflicting goals) are too strong. However, it provides a systematic way to react when global connectivity degrades, improving robustness in practice.

Chapter 6

Comparison of Strategies

6.1 Qualitative comparison

We briefly compare the three approaches: centralized, purely decentralized, and hybrid global–local.

Aspect	**Centralized**	**Decentralized**	**Hybrid global–local**
Decision variable	All u (global)	Local u_i per agent	Local u_i + global scalar gain
Constraint coupling	Fully global	Local neighbors	Local + global $\hat{\lambda}_2$ feedback
Computation	$O(N^2)$	$O(N)$	$O(N)$ + distributed estimation
Communication	Full network	Local neighborhoods	Local + $\hat{\lambda}_2$ consensus
Scalability	Limited	Excellent	Excellent
Global connectivity	Strong but costly	Only local	Enhanced via λ_2 feedback

Table 6.1: Comparison between centralized, decentralized, and hybrid global–local predicted-CBF control strategies.

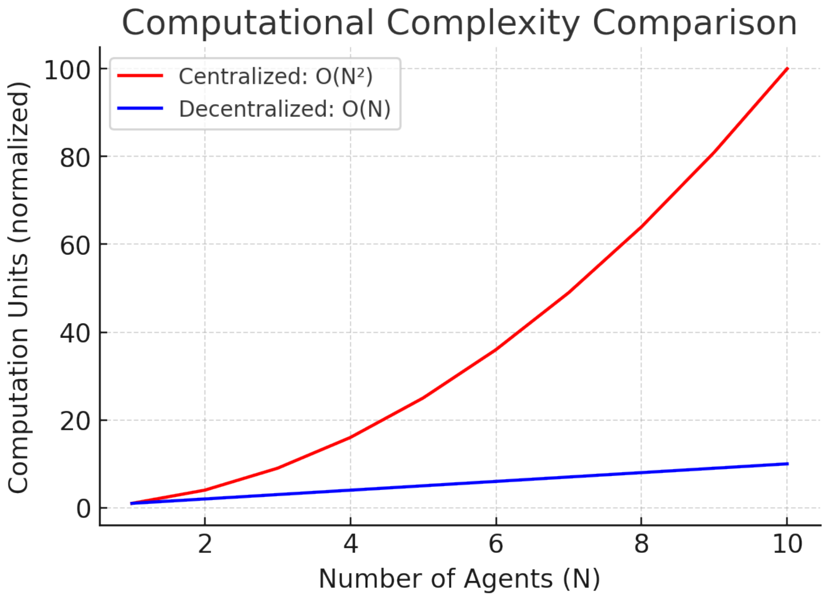

Figure 6.1: Comparison of computational and communication scalability: centralized ($O(N^2)$) vs decentralized/hybrid ($O(N)$).

Chapter 7

Conclusion

We have presented centralized, decentralized, and hybrid global–local CBF controllers for double-integrator multi-robot systems. The centralized approach yields globally consistent safety and connectivity at the cost of computational complexity. The decentralized approach scales well but relies on purely local constraints. The hybrid design introduces a λ_2-based connectivity feedback that modulates local CBF gains, enabling coherent swarm-level reactions when connectivity weakens.

These ideas can be extended with additional CLF constraints, formation objectives, and more sophisticated distributed estimators for λ_2 and its associated eigenvectors.